Rutherford Scattering

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Abstract

This lab report presents the findings of Rutherford's scattering experiment conducted at the Edwards Accelerator on the Ohio University campus. The experiment involved directing a ${}^{4}_{2}He^{2+}$ beam at a metal foil to study the scattering effects. The lab had three primary objectives: to perform energy calibration, confirm the composition of the foil, and compare theoretical and experimental differential cross-sections at various angles. The results indicated that the energy calibration produced residuals between -0.1 to 0.1 MeV, with a residual plot resembling a constant line. The foil composition was determined to be gold at $79.3 \pm 1.1\%$, silver at $10.99 \pm 0.65\%$, and copper at $9.7 \pm 1.3\%$. These values suggest that the measured composition fell within 0.1σ to 1σ of the desired proportions of 80%, 10%, and 10% for gold, silver, and copper, respectively. Regarding the theoretical and experimental cross-sections, only gold peaks were evaluated. Both exhibited a dependence on $\sin^{-4}(\theta/2)$, thereby confirming the validity of the Rutherford model after correcting systematics dead times. Furthermore, a ratio plot comparing the experimental and theoretical cross-sections was created for absolute angles. This plot indicated that the approximate ratio of all values converged around $R_{exp/theo} = 0.7$, revealing a discrepancy of only 30% between the experimental and theoretical cross-sections [1]. [Counts: 207]

1 Theory/Intro

1.1 Lab Overview/Topics

In this chapter of the lab report, an overview of the lab objectives and tasks will be presented, along with a discussion of the theory underlying the experiment. The class was divided into groups to perform offline analysis for Rutherford scattering data, focusing on varying energy or angles. The general overview is as follows: pre-run measurements of known quantities required for offline analysis were performed on the first day. On the second day, Rutherford scattering data was collected at varying angles. On the final day, data collection focused on varying energy[1]. The

concluding step involved offline analysis, including energy calibration, composition confirmation, and comparison between theoretical and experimental results. The experimental and analysis chapters will provide more detailed explanations of these steps. The remainder of this section will establish the theoretical foundations of Rutherford Scattering. Topics covered include explaining what a scattering experiment entails and introducing the statistical definitions of cross-sections and differential cross-sections. [Counts: 164]

1.2 Scattering/Cross-Section Theory

Before we define a differential cross-section, let's start defining a scattering experiment. A scattering experiment comprises two objects of interest in the whole system of said experiment: the incoming projectile beam and the target apparatus. The projectile beam can be anything you shoot toward the target, but in all but most cases for scattering experiments, it will always be composed of subatomic particles like alphas, electrons, any ion type, hadrons, etc. The target can be stationary or moving. This is based on the frame of the experiment and how it's designed, but I will get into that discussion in a bit. The target is also made up of groups of subatomic particles like the list discussed above. Still, stationary scattering experiments usually involve the nuclei of specific atom species, such as a gold nucleus. In this lab, the target combined the composition of gold, silver, and copper nuclei that made up the foil, and the projectile was a beam of alphas. As a projectile with its line of approach will either hard scatter with a much heavier target, which will then change the path of the incident beam at a certain point, the target-induced field (ex. Electromagnetic Repulsion) will eject the incident beam of particles and change its path this is what all scattering experiments are [2]. Scattering experiments are done so that the internal structure of atoms, interactions between projectile/target, and matter confirmation are proven/shown. A cross-section is essentially defined as the effective area of a target; here's the catch in most experiments, especially in subatomic collisions: the shape of the target is unknown [3]. A new definition of a cross-section has to be used, and it's actually defined as the statistical likelihood that a specific interaction occurs in an experiment with the incident beam and target. The interaction potential and statistics of the target/incident beam are what give the shape of the target. In the Rutherford Model, the potential that describes the process is the Coulomb Potential [3]. A differential cross-section is a probability that a specific interaction subtends the specific solid angle orientation of the target[3]. In this lab, the governing equation is defined as:

$$N_{\text{sc}} = N_{\text{inc}} n_{\text{tar}} \frac{d\sigma}{d\Omega}(\theta, \phi) d\Omega. \tag{1}$$

Specifically, Eq. 1 defines how to determine the differential cross-section experimentally given the incident particle beam count, number of scattering centers, target density, and solid angle. The Rutherford Cross-section in this lab is defined as:

$$\frac{d\sigma}{d\Omega} = 1.296 \left(\frac{Z_0 Z_1}{E_0}\right)^2 \left[\frac{1}{\sin^4\left(\frac{\theta}{2}\right)} - 2\left(\frac{M_0}{M_1}\right)^2\right] [\text{mb MeV}^2\ \text{sr}^{-1}]. \tag{2}$$

This version of the Rutherford Model is defined in nuclear physics units. Mass terms of index 0 and 1 represent the target and projectile masses, atomic number Z from both terms defines the

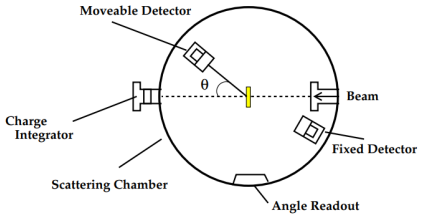
projectile and target charge, and E is the incident beam energy. Final thoughts from Eq. 2 are that the cross-sections are measured in the lab frame. [Counts: 483]

2 Equipment/Procedure

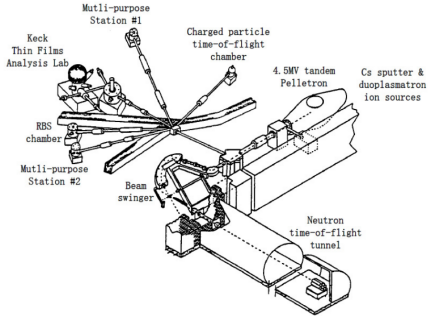
In this section, all the components used to run the experiment for each day run will be discussed, along with pre-run measurements, a description of trial run days, and how the raw data was displayed. [Counts: 34]

2.1 Equipment Pieces

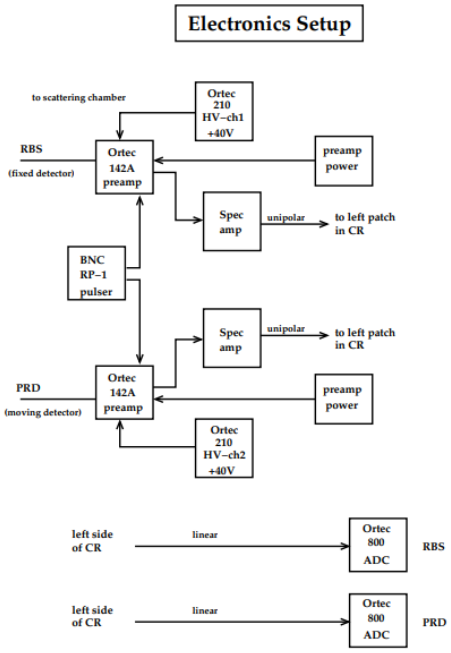
A quick overview of the experiment run will be addressed. The first thing was to have someone position the detectors at their corresponding angle settings using the angle readout settings in the vernier scale outside the chamber. This is shown in figure 1a. Then, the operator would set the energy settings for the alpha beam. The alpha beam is made from a Helium-4 isotope source called the duoplasmatron source [4]. As shown in figure 1b, the Helium-4 beam goes through a sequence of charge stripping/energy gain by the accelerator to become the specified alpha beam [4]. The alpha beam is directed toward the RBS station to begin the data collecting back-scattering and forward-scattering collection, which is shown in the DAQ system based on figure 1c. Other details to specify are that the charge integrator is what collects and specifies the total charge in the system, the scattering chamber encloses the station to prevent intermolecular reaction of air molecules and alphas, and the two detectors are what start the readout processes of taking an electrical signal to a digital pulse in a histogram/spectrum form [1]. [Counts: 185]



(a) Schematic of the Rutherford chamber for this experiment [1].



(b) Overview of the accelerator lab at Ohio University [4].



(c) Electrical setup showing how digitized counts are displayed [1].

Figure 1: Main components of this lab using the Edwards Facility at Ohio University to perform Rutherford Scattering experiments.

2.2 Pre-run Measurements

Figure 2 is a data collection table for solid angle calculations of the detectors. Figure 3 are the units used for this lab. Experimental differential cross-section units are defined by mb/sr. For the rest of this section, the convention of the detectors and data files is established. For both varying angle/energy data collection days, each data set is defined by a 4-tag.xy files, each ending with a number index. 000# are tag files for varying angle, and 100# for varying energy. the last digit that corresponds to 0,1, 2, and three in sequential order are the xy-files for the RBS, PRD, LT/DT, and BCI [5]. The only data files to use for offline analysis are the ones that correspond to RBS and PRD detectors. [Counts: 124]

Type	Name	Description
RBS	Distance	6.700 ± 0.035 cm
	Diameter	4.01 mm
	Angle (left)	$[225°, 335°]$
	Angle (right)	$[25°, 155°]$
PRD	Distance	6.075 ± 0.035 cm
	Diameter	3.69 mm
	Angle (fixed)	168°

Figure 2: Measurements taken from the RBS/PRD detectors and defining the angle ranges for the experiment [5].

$$1 \text{ barn} = 1 \text{ b} = 10^{-28} \text{ m}^2$$
$$1 \text{ millibarn} = 1 \text{ mb} = 10^{-31} \text{ m}^2$$

Figure 3: Units of area used in this lab [1].

2.3 Data Files to Process

Figures 4 and 5 are the data collected and organized from the DAQ system for varying angle and energy data ready to use for offline analysis. The final conventions to establish are that DTL is the dead times in percent form for the PRD detector and vice versa for the DTR association for the RBS detector. Other final notes for this section are that the fixed detector (RBS) was always established as $\theta = 168$ deg, and the movable (PRD) would vary or be set fixed based on a series of data runs. The detector angle is used as the scattering angle in this experiment, and essentially, if greater than $\theta > 90$ deg, the target is set to be $\theta = 0$ deg. If $\theta < 90$ deg, then the target is set to be half the detector angle. These are called back-scattering and forward-scattering processes. [Counts: 148]

Tag	$\theta_{\text{Det.}}$ [deg]	$\theta_{\text{targ.}}$ [deg]	I [nA]	BCI	DTL [%]	DTR [%]	Time [sec]
20	154.9	0	6	5000	1.0465	1.1628	86
30	60	30	6	5000	1.8391	1.3793	87
40	50	25	6	5000	5.4118	1.2941	85
50	40	20	6	5000	7.6136	1.25	88
60	-45 (215)	-22.5 (337.5)	6	5000	25.053	1.2629	95
70	-55 (305)	-27.5 (332.5)	6	5000	14.286	1.3187	91
80	-65 (295)	-32.5 (327.5)	6	5000	8.4043	1.276	94
90	-90 (270)	-45 (315)	6	5000	3.2673	1.2871	107
100	-90 (270)	-45 (315)	6	5000	3.8043	1.4130	92
110	-115 (245)	0	6	5000	1.5217	1.1957	92
120	-135 (225)	0	6	5000	1.1224	1.1224	98
130*	-35 (325)	-17.5 (342.5)	.6	5000	8.1818	0.56818	88
140*	30	15	.6	5000	10.28670	0.57143	105

(a) Data collection for varying angle data run [5].

Parameter	Value
E_{beam}	2.50 [MeV]
BCI Scale	10^{-8}
θ_{PRD}	168°

(b) Pre-definitions for angle varying data collection [5].

Figure 4: Collection of the tags, energies, scattering angles, BCI, and others for varying angle data collection [5].

Tag	E [MeV]	Θ_{RBS}	Θ_{Targ}	I [nA]	BCI	Scale [10^{-x}]	DTL [%]	DTR [%]	Time [sec]
1010	3.5	50.0 (1)	25.0 (5)	3	5000	8	1.9417	0.72816	206
1020	3.5	155.0 (1)	0.0 (1)	3	5000	8	0.46875	0.72917	192
1030	4.5	155.0 (1)	0.0 (1)	1.5	5000	8	0.34853	0.61662	373
1040	4.5	50.0 (1)	25.0 (1)	1.3	5000	8	0.96685	0.60773	373
1050*	6.5	50.0 (1)	25.0 (1)	1.5	10000	8	1.1679	0.024331	822
1060*	6.5	155.0 (1)	0.0 (1)	1.25	15000	8	0.027027	0.027027	1110

(a) Data collection for varying energy data run [5].

Type	Setting 1	Setting 2
θ_{PRD}	168°	168°
θ_{RBS}	50°	155°
θ_{Targ}	25°	0°

(b) Pre-definitions for energy varying data collection [5].

Figure 5: Collection of the tags, energies, scattering angles, BCI, and others for varying energy data collection [5].

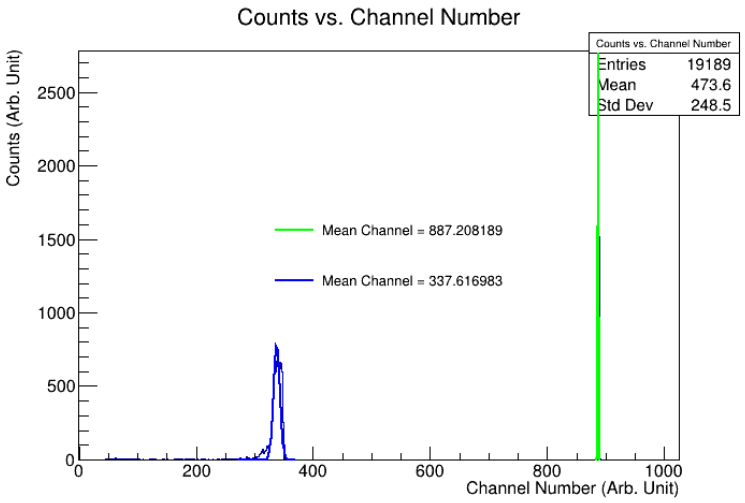
3 Analysis/Results

The offline analysis performed was energy calibration of the system, composition confirmation, and theoretical vs. experiment Rutherford. This section will mainly focus on offline analysis of varying angle data. A quick note on uncertainty standards and error propagation will also be discussed at the end of this section. [Counts: 47]

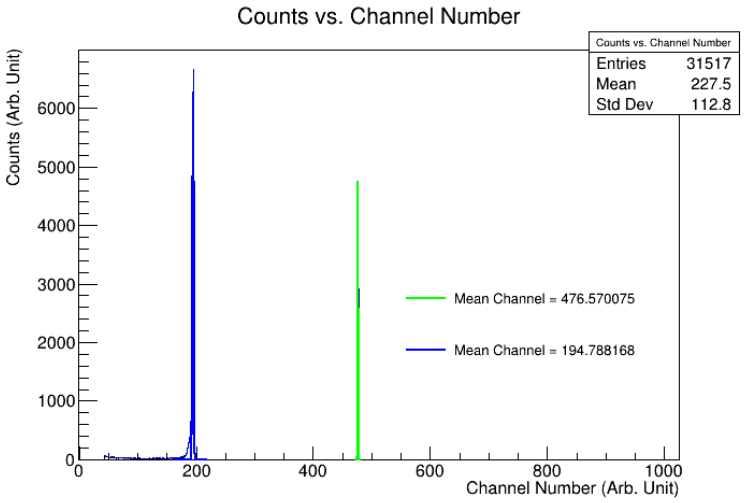
3.1 Energy Calibration

In the first steps in the offline analysis, energy calibration was performed on the given peaks. Energy calibration for this report was done on PRD gold peaks corresponding to when $\theta_{\text{Detector}} = 155, 50$ and energies ranging from 2.5-6.5 MeV. Those peaks were chosen because of the following: peak separation, identification, and known physics, which the experimenter can guess by loading up the histograms to see. The overall steps in calibration used were: spectra/parameter processing of xy-files into a histogram, scattered energy calculation of the chosen peaks, using graphing/analysis software like ROOT to then plot the corresponding calibration plot with fit parameters extracted and performing residual calculation and plot to verify the quality of calibration fit-function. At the end of this subsection and the beginning of the next section, it will show how to extract the fit-calibration equation to extract copper and silver peaks for their centroid position. [Counts: 148]

Peak Processing



(a) Example of organizing an RBS-based tag into a histogram and applying Gaussian fitting.



(b) Example of organizing a PRD-based tag into a histogram and applying Gaussian fitting.

Figure 6: RBS and PRD-based tags that are Gaussian fitted and processed. The sharp peaks are the pulsars shown in the distribution, and the peaks with the exponential tails are the corresponding gold, silver, and copper peaks. Gold is more prominent in these tag files.

Table of all Calibration Data/Step

Tags Used	Mean Bin	N_s (Counts)	E_{sc} (MeV)	E_{Cal} (MeV)	R (MeV)
tag0021	183.10 ± 0.04	9984 ± 16	2.283 ± 0.001	2.279 ± 0.001	0.004 ± 0.036
tag0041	196.00 ± 0.01	55917 ± 110	2.431 ± 0.001	2.432 ± 0.001	−0.001 ± 0.043
tag1011	280.42 ± 0.01	54386 ± 235	3.421 ± 0.002	3.429 ± 0.001	−0.008 ± 0.048
tag1021	262.46 ± 0.04	4836 ± 72	3.212 ± 0.001	3.217 ± 0.001	−0.005 ± 0.037
tag1031	340.91 ± 0.05	2897 ± 55	4.141 ± 0.001	4.143 ± 0.001	−0.002 ± 0.043
tag1041	364.02 ± 0.02	2381 ± 59	4.410 ± 0.002	4.416 ± 0.001	−0.006 ± 0.055
tag1051	529.78 ± 0.02	3136 ± 66	6.385 ± 0.003	6.373 ± 0.001	0.012 ± 0.069
tag1061	497.24 ± 0.04	4365 ± 67	5.995 ± 0.002	5.989 ± 0.001	0.006 ± 0.055

Table 1: This is all the data obtained from ROOT's fitting processes, developed algorithms, and plotting steps applied for the calibration analysis.

3.2 Composition Confirmation

3.3 Theory Vs. Experiment

Differential Cross Sections and Ratios Data

Tag Number	$\frac{d\sigma_{\text{theo}}}{d\Omega}$ (mb/sr)	$\frac{d\sigma_{\text{exp}}}{d\Omega}$ (mb/sr)	$R = \frac{\frac{d\sigma_{\text{exp}}}{d\Omega}}{\frac{d\sigma_{\text{theo}}}{d\Omega}}$
21	5698.0 ± 88.6	3619 ± 1265	0.635 ± 0.222
31	82820 ± 10015	5760 ± 2012	0.0695 ± 0.0257
41	162269 ± 24295	19216 ± 6713	0.118 ± 0.045
51	378292 ± 72561	20179 ± 7049	0.0533 ± 0.0213
61	241364 ± 40681	4719.9 ± 1649.2	0.0196 ± 0.0076
71	113868 ± 15271	74889 ± 26160	0.658 ± 0.246
81	62107 ± 6806	39528 ± 13808	0.636 ± 0.233
91	20702 ± 1446	12517 ± 4372	0.605 ± 0.215
101	10227 ± 455	4228 ± 1477	0.413 ± 0.146
111	7101 ± 205	4215 ± 1473	0.594 ± 0.208
121	633095 ± 140180	448482 ± 156663	0.708 ± 0.293
131	1153590 ± 300566	657537 ± 229687	0.570 ± 0.248

Table 2: Theoretical and Experimental Differential Cross Sections (mb/sr) and Ratios for 13 Tags of varying angle.

3.4 Uncertainties/Error Propagation Processes

4 Conclusion

Works Cited

[1] Dr. Paul King. *Physics 6751: Nuclear Lab Manual*. Version 24.5. Athens, Ohio, 2024.

[2] John R. Taylor. *Classical Mechanics*. Collisions, Central Force Problem. Sausalito, CA: University Science Books, 2005. Chap. 14, 8.

[3] Nouredine Zettili. *Quantum Mechanics: Concepts and Applications*. 2nd. Scattering Theory. Wiley, 2009. Chap. 11.

[4] Zach Meisel et al. “The Edwards Accelerator Laboratory at Ohio University”. In: *Physics Procedia* 90 (2017). Presented at the Conference on the Application of Accelerators in Research and Industry (CAARI 2016), Ft. Worth, TX, USA., pp. 448–454. DOI: 10.1016/j.phpro.2017.09.050.

[5] Ryan Conaway and Unish Gautam. *Rutherford: Lab Notes*. Lab Notes. Published on November 7, 2024. 2024. URL: N/A.